

PRADEEP KAVIRAJ

+91 9363623774 | pradeepmvk12@gmail.com | Bengaluru, India

github.com/PradeepKaviraj | linkedin.com/in/pradeep-kaviraj-283259324 | pradeepdev.duckdns.org

PROFESSIONAL SUMMARY

Full Stack Developer specializing in backend architecture, scalable systems, and AI integration within the JavaScript ecosystem (Next.js, Node.js, TypeScript). Proven ability to design REST APIs, implement secure authentication/authorization, and execute cloud (AWS) deployments. Strong focus on resolving complex engineering problems, API optimization, and maintaining real-world system stability from database architecture to production server management.

TECHNICAL SKILLS

- Backend & Architecture:** Node.js, REST APIs, System Design Basics, JWT Authentication, Role-Based Authorization, Middleware Architecture, Async Processing
- Frontend Development:** React.js, Next.js, TypeScript, JavaScript, Tailwind CSS, Framer Motion, Component Architecture
- Database & ORM:** MongoDB, MySQL, Prisma ORM, Relational Schema Design, Query Optimization
- Cloud & Deployment:** AWS EC2, Nginx Reverse Proxy, PM2, Vercel, SSL (Let's Encrypt), DNS Routing
- AI & Integration:** Large Language Models (LLaMA 3.1), OpenRouter API, Prompt Engineering, Streaming AI Responses
- Tools & Testing:** Git Version Control, Postman, Jest, API Testing, Error Handling

PROFESSIONAL EXPERIENCE

Software Developer Trainee | NexDesign Studio Pvt. Ltd., Bengaluru | Jan 2026 – Present

- Engineered modular React component architecture, standardizing UI patterns and eliminating codebase redundancy.
- Architected secure JWT-based authentication and authorization mechanisms to protect sensitive REST APIs and manage user access effectively.
- Designed scalable REST APIs and executed API optimization techniques on complex MongoDB queries, reducing data retrieval latency.
- Implemented comprehensive error handling protocols to debug and resolve complex full-stack integration issues, ensuring continuous application stability.

Software Developer Intern | NexDesign Studio Pvt. Ltd., Bengaluru | Jul 2025 – Dec 2025

- Developed production-ready full-stack web applications tailored for high-traffic fintech and educational platforms.
- Translated complex Figma system designs into responsive, high-performance user interfaces utilizing Next.js and Tailwind CSS.
- Orchestrated backend service integrations, implementing automated, asynchronous email notification workflows via Nodemailer.
- Maintained strict version control using Git workflows and configured continuous deployment pipelines via Vercel.

PROJECT ARCHITECTURE & ENGINEERING

AI SaaS Dashboard (Featured Platform) | Tech Stack: Next.js, TypeScript, Node.js, MongoDB, Prisma, OpenRouter API

- System Built:** Architected a full-stack AI platform supporting persistent, multi-thread user chat sessions.
- Backend Complexity:** Developed robust REST APIs to process real-time AI payloads, utilizing Prisma ORM for efficient MongoDB data modeling and indexing.

- **Architecture Decisions:** Designed a modular backend architecture enforcing strict JWT authentication, protected routing, and scalable user session storage.
- **Error Handling:** Engineered a comprehensive Jest testing suite (40+ test cases) validating API endpoints, database exception handling, and unauthorized access edge cases.
- **Real-World Relevance:** Deployed an enterprise-ready architecture capable of handling concurrent LLM processing and complex data persistence securely.

AI-Powered Developer Portfolio & Cloud Infrastructure | Tech Stack: Next.js, TypeScript, OpenRouter API, AWS EC2, Nginx, PM2, Linux

- **System Built:** Engineered a high-performance web portfolio featuring an integrated, stateful AI chatbot with real-time systems streaming.
- **Backend Complexity:** Integrated backend Nodemailer services and configured OpenRouter LLM APIs for seamless conversational responses.
- **Architecture Decisions:** Executed a bare-metal cloud deployment on AWS EC2 (Ubuntu), configuring Nginx as a reverse proxy to route traffic and PM2 for background process execution and auto-restarts.
- **Error Handling:** Implemented robust fallback UI states and API resilience mechanisms to prevent application crashes during LLM timeouts or network failures.
- **Real-World Relevance:** Secured the custom domain via DuckDNS using Let's Encrypt SSL/HTTPS and optimized server performance by actively managing Linux swap memory to resolve build bottlenecks.

Prison Management System | Tech Stack: React.js, Node.js, Express.js, MySQL

- **System Built:** Developed a comprehensive full-stack CRUD application for processing complex administrative and multi-module data.
- **Backend Complexity:** Built efficient Express.js REST APIs to handle high-volume data operations and secure data transit.
- **Architecture Decisions:** Designed a heavily normalized MySQL relational database schema to eliminate data redundancy and enforce referential integrity.

CLIENT PROJECT

Parents Choice (Insurance Platform)

- Engineered responsive frontend architectures and integrated backend asynchronous email communication workflows for user onboarding.

Creative Academy (Course Platform)

- Implemented a dynamic Next.js routing system and executed deep frontend optimization to support heavy content delivery without blocking the main thread.

EDUCATION

Bachelor of Computer Applications (BCA) SSMRV College, Bengaluru | 2022 – 2025

CGPA: 8.09 / 10

- **Core Focus:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems

CERTIFICATIONS

- **Full Stack Development** – NexDesign Studio
- **Advanced Diploma in Data Center Architecture**